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Review

Mixed farming and agroforestry systems: A systematic review on value chain implications

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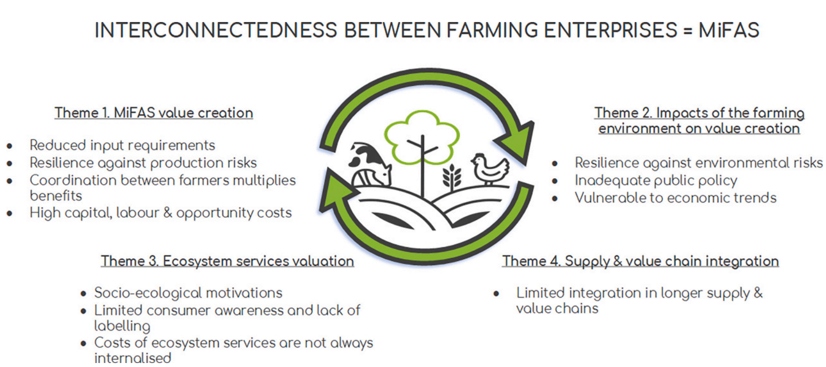
HIGHLIGHTS

- MiFAS may address ecological, economic, and social challenges, but insights into value chain implications are scarce.
- Based on a systematic literature review we find that inter-farmer partnerships can foster more cost-effective MiFAS.
- Collaborative enterprises for trading MiFAS (by-)products, processing, and sales, can strengthen MiFAS' economic prospects.
- MiFAS can potentially shorten supply chains and extend value capture for farmers by allowing greater ownership of value.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Mixed farming and agroforestry systems (MiFAS) are widely proposed to deal with ecological, economic, and social challenges that are associated with specialised farming systems. However, little is known about how MiFAS are integrated in food value chains and how value creation in MiFAS is rewarded by value chain actors.
OBJECTIVE: We review the broad literature on MiFAS (80 papers) with a particular focus on implications for food value chains.
METHODS: We use thematic analysis to code existing evidence and categorise these codes into four major themes: *MiFAS value creation, Impacts of the farming environment on value creation, Ecosystem service valuation, and Supply & value chain integration.* From here, we produce meta-narratives to summarise the literature within each theme. In a second step, we use these literature insights to develop possible value chain configurations that operationalise MiFAS in food value chains.
RESULTS AND CONCLUSIONS: We find that: a) integrating farming enterprises tends to improve yields and land-use efficiency and can reduce variable costs; b) MiFAS seem to face significant challenges with respect to fixed costs, such as of capital investments, and of labour; c) high opportunity costs associated with MiFAS can be a drawback for farms seeking to become more integrated as limited knowledge and the time to establish profitable MiFAS can deter adoption; d) sources of value creation within MiFAS stem from its ability to reduce variable costs as well as maintain or raise inputs, primarily by improving the efficiency of resources such as nutrients and

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land; and e) inter-farmer partnerships and collaboration have the potential to shorten MiFAS supply chains and extend value capture for farmers by allowing them greater ownership of value created.

SIGNIFICANCE: In total, insights into value chain implications of MiFAS are scarce. Therefore, our research delivers important implications for future research on MiFAS that should focus more on value chain implications of this emerging food production system.

1. Introduction

1.1. Introduction to mixed farming and agroforestry systems

The global agricultural paradigm is increasingly typified by specialised-intensive farming systems, characterised by their intensity of production, large scale, and specialisation (Lüscher et al., 2014; Moraine et al., 2014). Thanks to the advancement of machinery, agrochemicals, breeding programmes, and globalised supply chains, specialised systems are producing food in unprecedented quantities leading to the gradual diminishing of the threat of global hunger and malnourishment since the Green Revolution of the 1950s (Dalgaard et al., 2003; Evenson and Gollin, 2003). However, a growing scientific consensus charges specialised and intensive systems with the rapid depletion of nutrients in global soils (Borrelli et al., 2017), the excessive use of agrochemicals (Pingali, 2012), and increasing vulnerability to extreme weather (Olesen and Bindi, 2002). Economically, they are increasingly susceptible to price risks (Tothova, 2011) and are associated with longer supply chains with limited resilience capacities to external shocks (McCorriston and Sheldon, 2007; Meuwissen et al., 2019; Paas et al., 2021).

Mixed farming and agroforestry systems (MiFAS) present an opportunity to reduce some of the above symptoms. MiFAS emerged from the principles of agroecology, which aim to increase the sustainability and productivity of agriculture while maintaining the environment (Francis et al., 2003; Kremen and Miles, 2012). These systems seek to do so by incorporating scientific concepts from the disciplines of agronomy, ecology, sociology, and economics (Dalgaard et al., 2003; Gliessman, 2007). A MiFAS can be described as a farming system that integrates different farming and agroforestry enterprises (Fig. 1), wherein the multitude of processes and interactions between enterprises “create opportunities for synergistic resource transfers” over space and time

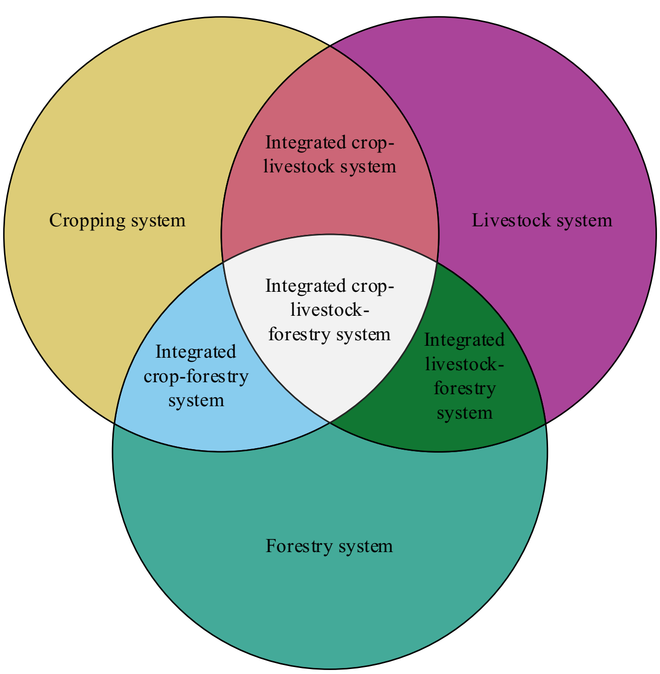

Fig. 1. Diagram depicting the different combinations of MiFAS, adapted from (EMBRAPA, 2016).

dimensions (Martin et al., 2016). MiFAS create synergies such that resources are more effectively utilised fostering more stable profits through diversification (Kirkegaard et al., 2014) and better environmental and ecological stewardship (Moraine et al., 2014; Ryschawy et al., 2012; Soussana and Lemaire, 2014). MiFAS have yielded promising results with respect to socio-economic (Bell et al., 2014; Darnhofer et al., 2010a; Havet et al., 2014; Peyraud et al., 2014; Wilkins, 2008) and environmental sustainability indicators (Duru et al., 2015; Hendrickson et al., 2008; Horlings and Marsden, 2011; Kremen and Miles, 2012; Power, 2010; Russelle et al., 2007; Ryschawy et al., 2012).

However, research explicitly addressing the value chain implications of employing MiFAS is missing, leading to uncertainty about MiFAS viability in the wider food production environment. This literature gap needs to be addressed to meet the following two issues. The limited value chain research is particularly anecdotal because it is disparate across different socio-economic regions, production systems and ecosystem processes, disciplines, and study designs. In a second issue, existing research has conceptualised MiFAS by fusing biological, agroecological, and – to an extent – economic processes with spatial, temporal, and organisational integration between farming enterprises (Martin et al., 2016; Moraine et al., 2014). While these can be sufficient for the farm-level, MiFAS have not been conceptualised to associate them with upstream and downstream value chains actors. To meet the literature gap, this review therefore has two main objectives: 1) to assess the potential value chain implications of MiFAS; and 2) to operationalise MiFAS configurations in order to facilitate the systematic and quantitative study of MiFAS integration in value chains.

1.2. Mixed farming and agroforestry value chains

The value chain is the chain of activities performed by an actor, each of which adds value to a product (Porter, 1998). These value-added activities represent the different processes that allow the value chain actor to create more value (whereas the supply chain concerns mainly the movements of products). In a MiFAS for instance, a farm may lower its operating costs by reducing the use of inputs such as agrochemicals and replacing them with lower cost inputs or alternative processes (e.g., no-till, manure-spreading), allowing it to improve profit margins (Garrett et al., 2017; Peyraud et al., 2014). Moreover, a farm could improve the qualities of its outputs by utilising more environmentally friendly practices that could raise the prices of its outputs and differentiate them from competing products. Total value is created along the so-called “long value chain” (totally incorporating the value chains of multiple actors), and products are sold on from actor to actor until the point of consumption (Nagurney, 2006). As products are sold, economic value is internalised by the selling actor (e.g., as profit), in simple terms reflecting market, and supply and demand forces.

Furthermore, some values are not always internalised, and are known as externalities. These can be negative (e.g., pollution) or positive (e.g., carbon capture). Externalities, the cost of producing negative externalities or the benefits of producing positive externalities, are not transmitted or valued economically along the value chain. As such, some processes and activities create value that is not in the form of a tangible output for revenue, particularly if they are ecologically or environmentally driven processes (Power, 2010). Being agroecological farming systems, MiFAS create value for the food production chain which can be more, or less, internalised, i.e., valued by the end consumers, other value chain actors, or public institutions. Such values can be environmental, ecological, or societal. For instance, labels are one such way to internalise externalities by appealing to consumers' willingness to pay them (Flinzberger et al., 2020; Röhrig et al., 2020). For our research, similarly to Gaitán-Cremaschi et al. (2015), we place the MiFAS in the long value chain where it creates and transmits tangible value while acknowledging that through ecosystem services produced it also creates additional externalities (Fig. 2).

2. Methods

2.1. Performing the literature search

To select articles that addressed the potential implications of MiFAS on value chains, we followed the PRISMA protocol (http://www.prisma-statement.org/) for performing systematic literature reviews (Moher et al., 2015). We formed a search string using the *Scopus* citation database (https://www.scopus.com) to search for relevant literature studying the value chain impacts of MiFAS. Logically, the search string was split into two sections joined by the AND operator so that articles returned in the search must meet two sets of inclusion criteria.

Keywords and terms in the first set were included to return papers on value or supply chains, farming system design, or which are explicitly review or framework papers. Acknowledging that value chain literature on MiFAS is sparse and our desire to keep a system-wide view (i.e., the whole MiFAS) rather than singular processes we selected broad-reaching keywords for this half of the search. We also specifically chose papers that include the root "compar*" (i.e., to return "comparison", or "comparing") for their breadth of scope

The second set was comprised of different synonyms and arrangements of MiFAS. These keywords were chosen based on our understanding of the various integrated farming systems encompassing a multiple of farming enterprises; namely arable, livestock, and forestry. We identified these, other synonyms and specific variations of MiFAS from our early scoping of the literature.

1st set:

TITLE-ABS-KEY ("supply chain" OR "value chain OR "farm* system design" OR "production chain" OR "upstream" OR "downstream") OR TITLE ("review" OR "framework" OR "compar*") OR KEY ("chain").

2nd set:

TITLE-ABS-KEY ("integrated agriculture" OR "integrated farm*" OR "mixed-farm*" OR "crop-livestock" OR "livestock-crop" OR "dual-purpose" OR "agroforestry" OR "crop-forestry" OR "forestry-crop" OR "livestock-forestry" OR "forestry-livestock" OR "silvopasture" OR "crop-livestock-forestry" OR ("crop" W/2 "forestry" AND "livestock") OR ("crop" W/2 "livestock" AND "forestry") OR ("livestock" W/2 "forestry" AND "crop") OR ("integrat*" W/2 "crop" OR "forestry" OR "livestock")).

Additionally, we specifically chose not to include keywords or terms relating to ecosystem services as much research has already been done about this topic in the service of agroecological farming systems. The valuation of ecosystem services also has its own strand of literature (for example see Hein et al., 2006; Sagoff, 2011). However, as we have acknowledged earlier (see Section 1.2), the nature of MiFAS as an agroecological farming system concerns processes that create societal and environmental value (or negate negative externalities). While not featuring in the search string, we nonetheless allowed in our methods and later thematic analysis (see Section 2.2) for the valuation of ecosystem services to emerge in our discussion of the literature, given its importance in the broader discussion of MiFAS value chains. We focused on peer-reviewed papers, namely articles and reviews published in journals indexed in *Scopus* at the time of the publication of this review and written in the English language. *Scopus* is a widely used database for scientific articles and includes articles from a large variety of academic fields and journals.

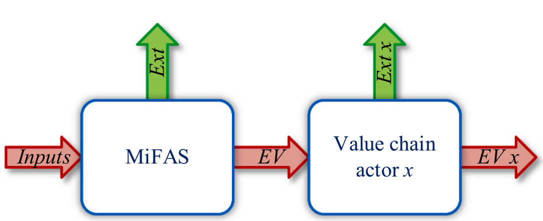

Fig. 2. Schematic representation of a MiFAS in the value chain, representing flows of economic value (*EV*) and externalities (*Ext*).

First, we performed a screening in order to eliminate duplicate entries and papers indexed on *Scopus* but for which no links to the articles could be found (e.g., .pdf files, journal hyperlinks, etc.) (Fig. 3). Following this, all papers were subjected to further screening to scrutinise their content and their relevancy. The three inclusion criteria were: 1) the paper is set in the developed economies of Europe (EU28), Northern America, Australasia, and Eastern Asia (excluding China); 2) the paper has a focus on agricultural and/or food production, or on agricultural supply and/or value chains; and 3) the paper can specify a mixed farming or agroforestry system. These three inclusion criteria were imposed for the retained papers to be topically and contextually relevant to the review. While developing economies present highly relevant and interesting studies on MiFAS (namely Brazil, China, and India), their supply and value chain contexts and the trends facing their agricultural systems offer paradigms too different from those found in the included regions. 80 papers were retained. It was noted during the screening that included papers varied greatly with respect to content, methodology, and relevance to value chain implications.

2.2. Methodology for analysing the literature

Thematic analysis is a method for identifying, analysing, and reporting patterns within qualitative data (Snyder, 2019; Ward et al., 2009). Specific approaches and methods for performing a systematic thematic analysis are established in Braun and Clarke (2006) and Thomas and Harden (2008), wherein qualitative codes (describing basic features of data) are generated from text and are then clustered into themes – the units for analysis. According to Braun and Clarke (2006), themes can be identified in two primary ways: through induction (bottom-up) or deduction (top-town). Our review used the deductive approach for the themes, mainly as we are concerned with exploring pre-determined themes. We identified four main themes centred around the creation and transmission of value in MiFAS chains.

The first principal theme was *MiFAS value creation*, centred upon value being created specifically within the farm-level boundary and considering the processes for value creation strictly performed within the MiFAS. Chiefly, these would be the values that are produced because of the integration of the various farming and agroforestry enterprises. The second theme was *Impacts of the farming environment on value creation*. This theme addressed the influencing factors that could influence value creation in the MiFAS, where the farming system must respond to the pressures outside of its control or boundary and which can affect how value is subsequently created by the MiFAS. Third was *Ecosystem services valuation*. This theme addressed the value that is not always immediately internalised through the sale of products from a MiFAS, where the value created and transmitted is often non-tangible, but which could be appraised economically, socially, and/or ecologically. Fourthly and lastly, the theme *Supply & value chain integration* concerned the integration of the MiFAS in the long value chain, where the value created by the farm-level is directly linked with the interactions that the farming system has with both upstream (e.g., for farming inputs) and downstream actors (e.g., buyers of farming outputs).

However, during our reading we identified in three of the four themes (*MiFAS value creation*, *Impacts of the farming environment on value creation*, and *Ecosystem services valuation*) that additional sub-themes were required to allow us to discuss complex sub-issues within themes (Braun and Clarke, 2006). In theme 1, *MiFAS value creation*, we

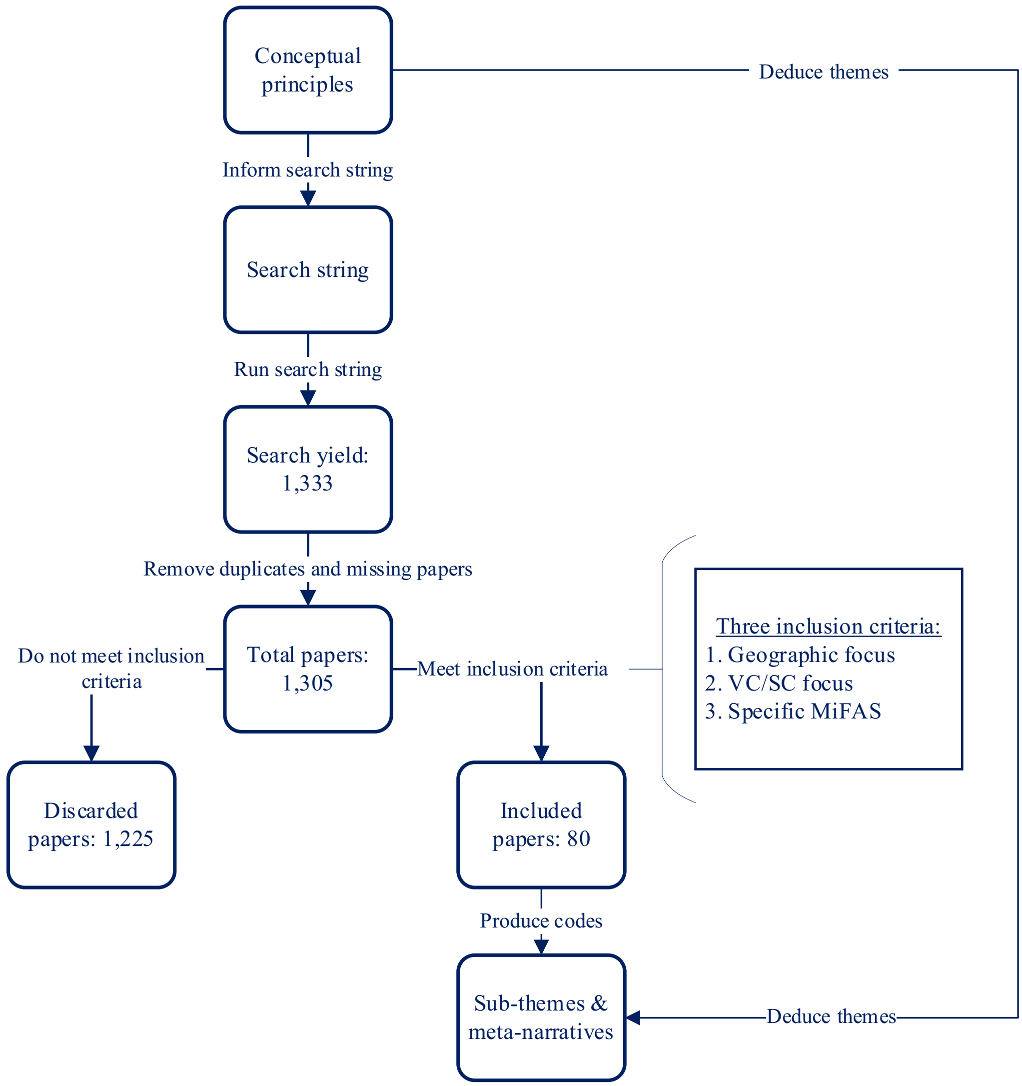

Fig. 3. Schematic overview of the method used to perform the literature review, following the PRISMA guidelines.

identified the two sub-themes *1a. Farm-level value chain* and *1b. Coordination management*. In theme 2, *Impacts of the farming environment on value creation* we identified three sub-themes: *2a. Biophysical environment*, *2b. Political & institutional environment*, and *2c. Economic environment*. In theme 3, *Ecosystem services valuation*, we identified two sub-themes: *3a. Economic valuation* and *3b. Ecological valuation*. Lastly, theme 4, *Supply & value chain integration* is standalone.

With the themes and sub-themes having been established the qualitative codes could then be produced from the literature. In each paper we noted whether it addressed a specific code. When a code was identified it was marked in an Excel spreadsheet (see supplementary spreadsheet), and each code was marked only once regardless of how prominent it was in a specific paper. The codes were categorised under the sub-themes to extract the meta-narratives (see section 3.2). The extraction of meta-narratives brings together the storylines within each theme. As MiFAS literature is diverse and specific research on value chains is limited, we produced meta-narratives in order to qualitatively review thematically relevant literature (Snyder, 2019). Meta-narratives are used to tell a compelling and coherent story from the literature while providing concrete evidence and examples from the data to "capture the essence of the point you [*sic*] are demonstrating, without unnecessary complexity" (Braun and Clarke, 2006). Meta-narratives are comparable with meta-analyses, where the latter is typically applied to summarise quantitative empirical literature (DerSimonian and Laird, 1986; Wong et al., 2013).

2.3. Conceptualising MiFAS configurations

While analysing the literature, we observed that a common understanding of how MiFAS value chain actors interact with each other is lacking. Therefore, we conceptualised MiFAS value chain configurations informed by what we had observed in the papers being included in our thematic analysis. We were suitably placed to identify how different configurations of MiFAS take shape and believed that an overview of these could help to identify and refer to MiFAS configurations in future research on value chains.

3. Results

3.1. Descriptive overview of the literature review

Our search yielded 1333 results (Fig. 3). The first screening led to the exclusion of 28 duplicate or missing papers and 80 papers were retained for the thematic analysis. We noted during the screening that the retained papers varied greatly in content, MiFAS types, methodology, and relevance to value chain implications. 62 papers, representing over 77% of the total, were published since 2012 (last 10 years). 50 papers were indexed in *Scopus* as "Article" and 30 as "Review" (see supplementary spreadsheetfor the full list of included papers).

We identified 42 distinct codes which we categorised under the four themes (see supplementary table A1 for the code list). Codes in sub-theme *1a. Farm-level value chain* were the most frequently occurring (198 times), while also being the most numerous amongst all sub-themes (14 codes) (Fig. 4). On the other hand, theme 4. *Supply & value chain integration* had both the fewest code occurrences and number of codes, at 16 and 2 respectively. In total, the plurality of papers discussed *MiFAS value creation* above all other themes. Codes in sub-theme *3b.* were on average the most frequently cited, meanwhile sub-theme *3a.* had the fewest mean code citations. Even within sub-themes, some codes were particularly prominent, such as code 5: *Ecosystem services in MiFAS decrease variable input usage and costs* (sub-theme *1a.*) and code 39: *Socio-economic value-added motivates MiFAS* (sub-theme *3a.*), suggesting that some research is niched along specific discussion points. It is evident that some sub-themes represent significant research fields, while others are less explored – namely concerning farm business economics and the value chain. We can also see (in supplementary spreadsheet) that some papers address specific sub-themes. For example, the paper by Flinzberger et al. (2020) is particularly concerned with product labelling in agroforestry landscapes and makes significant contributions to the meta-narrative of sub-theme *3a*. Other papers, particularly reviews, often transect (sub-)themes. Therefore, it is important to consider the impact that individual papers had on the meta-narratives.

3.2. Themes & meta-narratives

3.2.1. MiFAS value creation

3.2.1.1. Sub-theme 1a. Farm-level value chain. Sub-theme *Farm-level value chain* emerged as a distinct sub-theme as its primary focus was on the processes performed strictly through the integration and interconnectedness of farming enterprises within the so-called "farming boundary"; i.e., farming activities. This entails value being produced through the integration of production of the various farming and agroforestry enterprises over time and space dimensions.

It was found that integrating farming enterprises tends to improve yields and land-use efficiency and can reduce variable costs. Namely, most of these benefits are tied directly to the provision of ecosystem services, which decrease the need for synthetic inputs such as fertilisers and pesticides (being replaced instead by biological processes) and therefore can reduce variable costs, and can improve yields particularly by allowing for synergistic and continuous production cycles that allow land to be used more optimally and mobilise greater biological activity. Several papers suggest that the land equivalent ratios of MiFAS are the same or exceed those of specialised systems. Graves et al. (2007) modelled land equivalent ratios in European integrated crop-forestry systems, and found them to be between 1 and 1.4, indicating that, under typical management, integrating crops and trees on the same land is more productive. Furthermore, the functions of ecosystem services are reinforced by time and a greater diversity of biological processes in the farming system up to and including pest and weed suppression (Malézieux et al., 2009) and enhanced soil fertility (Tsonkova et al., 2012). Therefore, as ecosystem services are continually provisioned within the farming system the greater the purported production benefits. As such, the benefits should be evaluated in the middle or long term as systems and processes require time to mature.

However, increases in yields and efficiency can be withheld by constraints. Firstly, land may be used more effectively, but only up to the point where it is not being used competitively amongst farming activities. For ecosystem services to materialise into production benefits, diversification should lead to synergies. In such cases, poor management of MiFAS can undo benefits. For example, tree shade in agroforestry is an important management consideration when integrating them with crops (Tsonkova et al., 2012) and livestock (Anderson and Batini, 1984). Additionally, animal droppings may not be evenly distributed across pastures in crop-livestock rotations overloading some areas while leaving others deficient (Sekaran et al., 2021). Secondly, while variable costs can be reduced, MiFAS also seem to face significant challenges with respect to fixed costs, such as of capital investments, and of labour. Diversified holdings tend to require more farming infrastructure and machinery and of labour (likely exceeding what can be available on a family farm, for instance); as such MiFAS may also be harder to upscale due to these cost constraints. Correspondingly, the high opportunity costs associated with MiFAS can also be a drawback for farms seeking to become more integrated as limited knowledge and the time to establish profitable MiFAS can deter adoption. Marton et al. (2016) suggest that specialised crop farms have higher yields than mixed crop-livestock farms leading to higher phosphorous use efficiency (while crop-livestock farms were more nitrogen and potassium efficient), and that this could be attributed to better crop management practices: a crop-livestock farmer needs to be a generalist with knowledge about cropping and livestock, while a specialised farmer needs to only focus on one activity.

While the effects of integration on costs seem mixed, MiFAS present a greater degree of flexibility against production risks and can perform better than specialised systems under adversity, such as by providing income stability as diversification can spread revenue streams (Garrett et al., 2017). Ecosystem functions also seem to play a reinforcing role in increasing resilience capacities and the longer a farm remains integrated and diversified, the more such a farm can manage its resilience to adverse conditions (Peterson et al., 2020; Thiessen Martens et al., 2015).

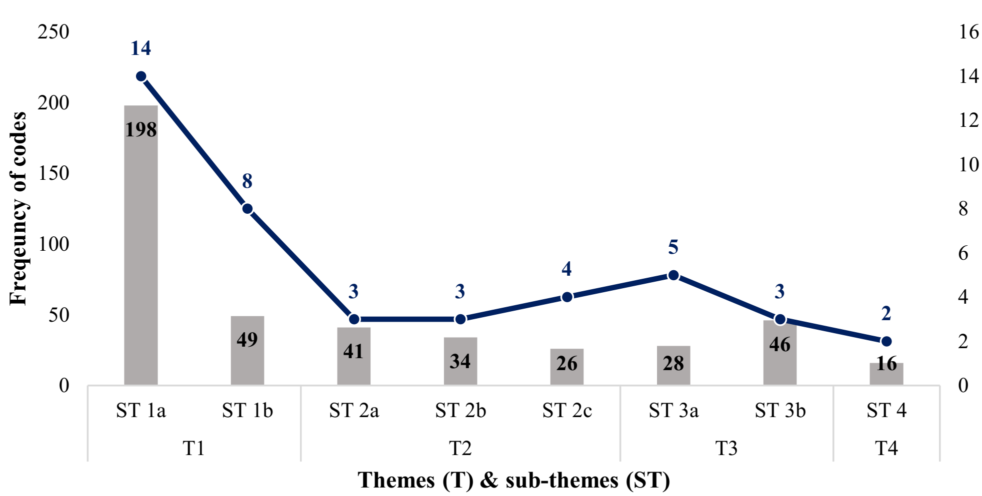

Fig. 4. Summary of the results: frequency (represented by columns and the left *x* axis) and number (represented by the line and the right *x* axis) of codes by theme (T) and sub-theme (ST). T1 = *MiFAS value creation*, ST 1a = *Farm-level value chain*, ST 1b = *Coordination management*; T2 = *Impacts of the farming environment on value creation*, ST 2a = *Biophysical environment*, ST 2b = *Political & institutional environment*, ST 2c = *Economic environment*; T3 = *Ecosystem services valuation*, ST 3a = *Economic valuation*, ST 3b = *Ecological valuation*; T4 & ST 4 = *Supply & value chain integration.*

In essence, sources of value creation within a MiFAS stem from its ability to reduce variable costs as well as maintain if not raise outputs, primarily by improving the efficiency that resources such as nutrients and land are utilised. However, value creation in a MiFAS is stymied by the high cost to work an integrated system in terms of working and human capital. Although MiFAS requires a great investment, the specific advantages of MiFAS versus specialised or less integrated/diversified farming systems may manifest in their ability to be more resilient under adverse conditions and over longer periods of time. Rather than assessing the added value of MiFAS in the short term, the benefits of MiFAS may need a more nuanced, longer-term perspective and assessment.

3.2.1.2. Sub-theme 1b. Coordination management. Distinctly from *Farm-level value chain*, this sub-theme emerged to address the importance of a third dimension of integration: coordination management (recalling that integration also takes place over space and time dimensions, see Section 2.1). Particularly, the management of integrative processes was identified as a key driver for their intensity and scale; where the management capabilities, capacities, and desires of farmers may affect the degree to which the farming system is integrative. Additionally, this allowed us to interpret a MiFAS farming boundary as one that can also enclose multiple interconnected farm businesses, including functionally specialised farms.

It was found that increased coordination between farmers is highly desirable, as it implicitly allows for greater integration between the enterprises themselves. In so doing, coordination between farm managers enables the provision of more ecosystem services (Martin et al., 2016; Moraine et al., 2017), which as mentioned earlier has a positive knock-on effect on production synergy, while additionally enabling internal markets and supply chains to form allowing farms to improve value creation and retention (Röhrig et al., 2020) potentially increasing the farming system's ability to negotiate with downstream actors. Internal markets themselves can also be partly facilitated through the standardisation of production (Moraine et al., 2014; Ryschawy et al., 2019). With closer integration, value may also be added into the MiFAS boundary by reducing costs of labour and opportunity costs of knowledge through the pooling of resources (Moraine et al., 2014; Peyraud et al., 2014), while simultaneously permitting greater risk sharing and flexibility against adversity.

However, increased coordination and integration between farms has been found to be difficult, as synergising processes and operations over time and space requires extensive management from involved parties (Alary et al., 2019; Hilimire, 2011; Martin et al., 2016; Ryschawy et al., 2019); differences in management styles, planning, risk attitudes, conflicting interests, and reduced individual ownership of processes only become more exacerbated as the prerequisites (such as trust) and costs for closer integration increase. For example, (Moraine et al., 2016) describe how, by engaging with each other through cooperative structures, farmers may lose control over the management and governance of their activities.

By expanding the definition for what a MiFAS boundary may encompass, i.e., multiple farm businesses, we could identify that coordination between farms has the potential to multiply the benefits of integration at the farm-level, such as by increasing the provision of ecosystem services and reducing operating costs. Furthermore, it may also enable farms with different specialisations to engage with one another and make use of comparative advantages, such as in knowledge and capital. Internal markets and supply chains facilitate trade and exchange of by-products and resources such as land. Significantly, this may also enable farms to increase the territory upon which integrative operations take place for instance, expanding the useable area for farming rotations. However, the high management cost of coordination hinders the adoption of more integrative practices between farming enterprises; interpersonal challenges may not necessarily be overcome by economic arguments alone. It may suffice that integrating practices may be kept to an acceptable maximum while the source of value-added comes from the system's interactions with downstream value chain actors.

3.2.2. Impacts of the farming environment on value creation

3.2.2.1. Sub-theme 2a. Biophysical environment. Darnhofer et al. (2010b) indicated that the behaviour of farming systems is nested within three main domains: the ecological (for our sake named biophysical); policies and social norms (political and institutional); and economic. We purposed these domains as broadly concerned with factors that can influence value creation in a MiFAS, where the farming system must respond to pressures outside of its control or boundary and which can affect how value is subsequently created by the MiFAS.

Sub-theme *Biophysical environment* identified three codes. Agro-climatic conditions such as weather and geography may constrain the kinds of ecosystem services produced and forms of integration. In the high rainfall zone of southern Australia, (Nie et al., 2016) describe that in drought events crop-pasture intercropping (rotating cropping and livestock) may exacerbate grain yield penalties as a result of competition for above- and below-ground resources, while stubble grazing by livestock may reduce soil cover essential for moisture retention and organic matter recycling. It should be noted that in Nie et al. (2016), yields can be improved under normal growing conditions. On the other extreme, Alary et al. (2019) describe that if climatic and growing conditions are ideal for specific agricultural productions, lower-value productions are pushed towards more marginal and less productive lands enforcing agricultural specialisation in regions. For example, in southern France, areas suitable for high-value permacultures (such as vineyards and orchards) tend to exclude livestock (Alary et al., 2019). As such, growing conditions may cause the dislocation and disassociation of farming enterprises and limit integration.

On the upside, MiFAS may nonetheless take better advantage of marginal agricultural lands, where the provision of ecosystem services may enhance the suitability and profitability of farming enterprises. In New Zealand, the integration of trees (including for timber) in pasturelands on North Island alleviates issues of soil erosion arising from difficult topography, high rainfall and highly erodible soils, enabling farmers to maintain productive systems in lands sometimes even too steep for livestock systems alone (Cubbage et al., 2012). The adoption of silvopasture and silvoarable systems in areas exhibiting high temperature and humidity conditions such as in the southern United States can both improve the productivity and wellbeing of livestock and reduce the severity of crop-losses due to drought and flooding, respectively (Cubbage et al., 2012). By extension, as climate and weather risks become more frequent and severe due to the onset of climate change, some MiFAS have also been shown to be more resilient. Integrated livestock-forestry systems such as the *montado* or *dehesa* in Iberia are less prone to the risk of wildfires as a result of managed grazing of combustible materials while also providing shade to animals (Aguilera et al., 2020; Ortega et al., 2012). Increased dryland soil salinity in Australia forced on by land-use change and land and water degradation could be offset by productive agroforestry (George et al., 2012).

We found that the biophysical environment plays a particularly important role in determining the kinds and amount of ecosystem services produced by a MiFAS. While it is found that some integrations or MiFAS are not necessarily optimal under specific environmental conditions, or when such a situation arises where the value of ecosystem services are not greater (in terms perhaps of economic returns) than employing specialised systems, MiFAS may nonetheless be better suited to utilise marginal lands where high-value production is limited. In cases where the provision of ecosystem services through integration improves the productivity and resilience of farming systems over specialised systems, a MiFAS may be able to create more value-added, particularly over longer periods of time where changes to weather patterns are concerned.

This could open the possibility for repurposing land that has otherwise been abandoned due to environmental and geophysical challenges with relatively higher value-added productions.

3.2.2.2. Sub-theme 2b. Political & institutional environment. Under sub-theme *Political & institutional environment*, we identified that significant barriers exist for the adoption or practice of MiFAS. We find that policy has a tendency towards disincentivising MiFAS. In California, for example, the *Leafy Green Marketing Agreement* regulation requires the passing of one year between the application of raw manure and the planting of leafy greens, effectively forcing farmers to separate crops from livestock in order to comply. While meeting the regulation is voluntary, downstream buyers may identify produce coming from integrated crop-livestock systems as hazardous due to potential food contamination (Hilimire, 2011). In another case, Reeg (2011) describes that farmers in Germany may be uncertain about future policy changes that affect long-term land usage such as the planting of trees. Barriers can emerge from a lack of subsidies, support or awareness of the ecological and societal value that MiFAS may have; in the EU, for example, there is a distinct lack of subsidies or support schemes integrated in the Common Agricultural Policy for MiFAS, affecting economic profitability, causing land-use polarisation, and harming the integrity of existing integrated systems (Flinzberger et al., 2020; Havet et al., 2014). All the while, agricultural policies in developed regions (such as in Europe) have historically focused on increasing productivity, increasing health safety and standardising agricultural production. In recent years awareness for the impacts that agriculture has on the environment has led to a gradual shift in policy objectives (Graves et al., 2007). Farmers increasingly need to comply with environmental regulations or are incentivised to take agro-environmental measures to address the negative impacts of agriculture (such as in the EU's Common Agricultural Policy), though they largely remain beholden to regional and international markets for economic viability (Duru and Therond, 2015). As a result of weak policy, few farmers are incentivised to either transition to or maintain MiFAS, and this has the knock-on effect of reinforcing the specialisation of farming systems, leading to an institutional lock-in effect of both management skill specialisation and supply chain specialisation (Duru and Therond, 2015; Garrett et al., 2017).

3.2.2.3. Sub-theme 2c. Economic environment. In this sub-theme, the most significant issue identified centred around the high cost of labour, which more significantly affects farming systems that perform labour intensive activities such as those involved in integrated farming. Labour requirements to perform integrated farming tend to increase or become more intense, offsetting cost reductions. Exacerbating this issue is the lack of social capital in farming regions in the form of rural abandonment (Martin et al., 2016) and the lack of generational succession (Ryschawy et al., 2014), leading to labour shortages that could severely affect MiFAS. For instance, an evaluation of crop-livestock systems in the United States suggested that labour requirements increased by 59% and 232% when alfalfa and livestock are introduced to farming systems, respectively, while at the same time an aging and diminishing rural population makes labour intensive work less sustainable (Hendrickson, 2020), forcing systems to specialise (requiring less labour and capital). In other regions, such as in the Mediterranean, high costs and low profitability of extensive livestock systems have further led to rural abandonment and intensification, at the expense of extensive crop-livestock and agroforestry systems that require both ecological diversity and knowledge capital (Aguilera et al., 2020). However, in being more diversified, MiFAS can nonetheless be more resilient against market and price risks such as price volatility for inputs and outputs; the former due to being less dependent on external inputs, and the latter due to diversification (Malézieux et al., 2009; Monge et al., 2016).

It emerged that specialised farming systems tended to be both more affordable and straightforward than MiFAS, being less dependent on increasingly limited socio-economic capital of agricultural regions. While market and price risks may be reduced in MiFAS, a significant limiting factor surrounding the viability of MiFAS seemed to concern how well they can be integrated in farming regions where broader socio-economic trends are pushing towards specialisation. Interestingly, the literature has suggested that social vulnerability, for instance in rural job security, could proxy a farming system's ability to cope socio-economically to adverse conditions (Sneessens et al., 2019); while labour demands may increase in MiFAS, such farming systems that require more fulfilling and permanent employment could alleviate trends in depopulating rural areas.

3.2.3. Ecosystem services valuation

3.2.3.1. Sub-theme 3a. Economic valuation. Sub-theme *Economic valuation* emerged as the first of two sub-themes on ecosystem services valuation. We recognised that value is not always immediately internalised through the sale of MiFAS products where value created and transmitted is sometimes intangible, but which could be appraised in the value chain. In this regard, the literature has highlighted labelling as a positive recourse towards capturing more value within the MiFAS boundary, as labelling outputs that are produced using ecosystem services increases consumers' perceived value. For instance, labelling and certification schemes permit quality assurance to consumers (Röhrig et al., 2020). Additionally, labelling outputs from regions traditionally employing MiFAS, such as with geographic indicators, may also support their conservation by increasing value-added and alleviating issues of land abandonment and intensification (Flinzberger et al., 2020). Labels may also support the establishment of internal markets that can strengthen the cohesion and social interactions of integrated farmers (Moraine et al., 2014). Important social implications may positively value MiFAS, such as enhanced landscape beauty and wildlife diversity that can foster tourism, recreation and hunting activities (Le Houérou, 1993); arising cultural functions and employment opportunities add further value to the system (Lovell et al., 2010).

Barriers against labelling exist, however. Firstly, labelling and certification standards are simply lacking. This can be due in part to the costs of producing, implementing, and marketing a label being too expensive, all the while market infrastructure and consumer awareness of the value-added of MiFAS products are negligible. Implicit in the labelling process requires a substantial degree of coordination and compliance between MiFAS, in order to both market outputs and reduce costs. Cooperative structures and collectively processing primary outputs into added-value secondary products allow farmers to take greater ownership of their value chain and capture more value. It was suggested in Röhrig et al. (2020) that small scale, direct marketing of MiFAS products could both overcome marketing and labelling costs by appealing to consumers' beliefs of nutritional and physical characteristics of products, taste, origin, animal welfare and environmental stewardship.

3.2.3.2. Sub-theme 3b. Ecological valuation. Ecosystem services may alternatively create ecological value that is not appraised in the value chain and instead produce externalities that are nonetheless consumed in some fashion. Valuing the ecological externalities produced from ecosystem services should have for an objective to make producing positive externalities more equitable. Conservationism seems to be a motivator for implementing and supporting MiFAS, as such systems are dependent and interlinked with the local socio-ecologies that they are situated in (Pavlidis and Tsihrintzis, 2018), such as the *montado* system that has played an important historic role in the Portuguese agricultural landscape (Flinzberger et al., 2020). Beneficiaries of the impacts of the ecosystem services produced by MiFAS (such as in reducing negative externalities and creating positive externalities) tend to be both the farmer and the public domain. The farmer may benefit from ecosystem services by increasing their resilience and reducing natural and ecological risks (such as against weather shocks, fire incidence or disease outbreaks). The public domain broadly benefits from better agricultural practices that may limit environmental and ecological degradation, for example (Alary et al., 2019; Campos et al., 2020). However, the need to form mechanisms to support the provision of these ecosystem services must come through better evaluating the true cost of producing them. Subsidies or environmental credit schemes may help to promote MiFAS, as would appraising the intermediate processes that produce inputs to be consumed during the production process of primary outputs; for instance, inputs that would otherwise be considered without value such as by-products or waste (e.g., rotten fruit or overgrowth that can be eaten by livestock) (Campos et al., 2020).

3.2.4. Supply & value chain integration

This standalone theme concerns the integration of MiFAS in the long value chain, where the value created by the farm-level is directly linked with the interactions the farming system has with both upstream actors (e.g., for farming inputs) and downstream actors (e.g., buyers of farming outputs). It was found that weak supply and production chain integration and influence reduces that amount of value captured by MiFAS. For example, farms participating in long supply chains in globalised markets need to concentrate and specialise production in order to be competitive (Moraine et al., 2014). Farms that are diversified and reduce their use of external inputs also participate less in their procurement from the market (taking less advantage of economy of scale, though in return potentially making up for it in economy of scope) (Havet et al., 2014). Even integration at a territorial level may present additional challenges in the form of additional transportation and logistical costs and need for

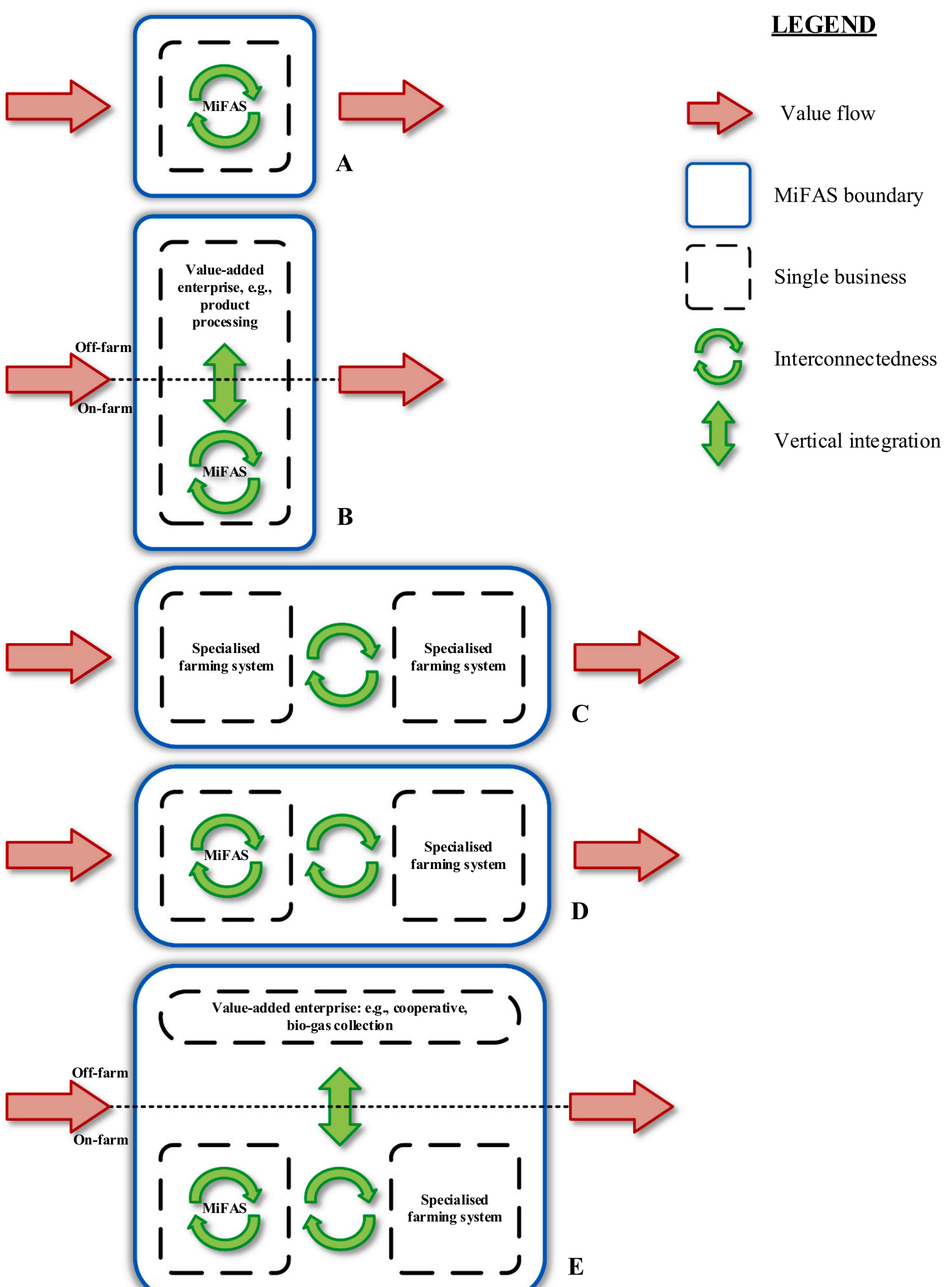

Fig. 5. A = Single-farm MiFAS, B = Single-farm integrated MiFAS, C = Multi-farm landscape MiFAS, D = Multi-farm hybrid landscape MiFAS, E = Multi-farm integrated landscape MiFAS.

increased management and organisation (Garrett et al., 2017; Moraine et al., 2014) where physical infrastructure and distance may impede some territorial MiFAS. Shortening the supply chain and long value chain in order to reach sellers of inputs and buyer of outputs may help to alleviate some stresses. The local procurement of inputs can reduce logistical costs, where closer integration between farmers may lead to Coasean agreements that extend value captured within MiFAS boundaries (Havet et al., 2014; Röhrig et al., 2020). Likewise, direct marketing, voluntary price signalling, and on-site processing may increase the perceived value of products and place farmers in MiFAS closer to the final consumers of their products, reducing how much value is passed to intermediary value chain actors and enabling farmers to capture more value-added (Alam et al., 2014; Röhrig et al., 2020). For instance, once farmers are able to communicate directly with consumers or through local sales channels about their products and product qualities, consumers are more receptive to ecosystem services in product value (Lovell et al., 2010; Röhrig et al., 2020).

3.3. Mixed farming and agroforestry system configurations

While the meta-narratives were being produced, we understood that there was a lack of a framework available for the study of MiFAS, particularly with respect to how value is created. All the while, the literature tended to focus on the merits of specific interspecies interactions or processes, and rarely with the viewpoint of value creation by the farming system. In the process of producing the meta-narratives, we became increasingly informed of the means by which value is created within the MiFAS boundary (i.e., *MiFAS value creation*); how different external factors can influence value creation in the MiFAS (i.e., *Impacts of the farming environment on value creation*); how the externalities created by a MiFAS are or could potentially be appraised in the long value chain (i.e., *Ecosystem services valuation*), and; how MiFAS are integrated in wider supply and value chains (i.e., *Supply & value chain integration*).

As a result, we propose the following Figs. 5a-5e. as baseline MiFAS configurations. We hypothesise that each configuration creates, captures and transmits value differently. Each configuration is designed such that a researcher may more concretely describe the following: the value-added processes performed within the MiFAS boundary; and the placement of and connections between value chain actors both within the boundary and without. The configurations themselves are universal, not focusing on the individual arrangements of species or processes but instead on the characteristics and standards of the systems they represent. Integration between farming enterprises is referred to in these configurations as “interconnectedness” in order to encapsulate synergies as well as drawbacks.

Each configuration is represented by a MiFAS boundary that encloses the sum of all the operations that take place by the actors directly involved and managing the interconnectivity of farming enterprises. These actors are principally the farmers and the beneficiaries of the MiFAS. The economic value created within the MiFAS boundary is the sum of the value of outputs leaving the boundary (revenue) less the sum of the value of inputs entering the boundary (costs). From Porter (1998), this difference encapsulates the primary activities performed by the boundary: inbound logistics, operations, outbound logistics, marketing and sales, and service; and the secondary activities: firm infrastructure, human resource management, technology development, and procurement. As such, value created and captured in the MiFAS boundary is tangible, possessing economic value. In our configurations we also allow for cash subsidies to also factor within the MiFAS boundary. Additionally, the configurations account for value flows of ecosystem services that are internalised including internally consumed ecosystem services, though not externalities (which are not consumed).

In Fig. 5A the above is simply represented as a MiFAS boundary that encloses a farming business, a single farm holding. Within this single business the processes of integrating farming enterprises takes place. The value created by the MiFAS is retained within the boundary and – because there is one actor in this boundary – captured by the farm business. Fig. 5B includes the addition of another value-added enterprise that is performed by the farm business but which itself is not a farming activity. Such an enterprise can be processing dairy into cheese or fruit into jam, et cetera to create refined outputs for later sale. This could also include providing services that create value, such as touristic services. In Fig. 5B we denote the implicit connection between the interconnectedness of the farming enterprises and the off-farm enterprises as being vertically integrated, where performing additional non-farming related activities (though directly depending on them) extends the value captured by the system boundary.

In Fig. 5C the boundary now encloses two or more specialised farming businesses where the interconnectedness between the farming actors makes up the MiFAS boundary. As in the previous configurations the value creation processes from integrating farming enterprises are much the same, though they now transect the landscape or farming territory. In effect, there are multiple value chains existing in the boundary belonging to the various farm businesses; the amount and depth of coordination between them raises the shared value-added. Martin et al. (2016) distinguish four types of spatial, temporal and organisational coordination options between farms: global coexistence, local coexistence, complementarity and synergy. Under global coexistence no coordination takes place between farms and thus the technical and organisational options for farms are market driven. In the remaining three types, increasingly closer coordination instigates additional activities and processes that are performed by farms within the boundary. Fig. 5D builds on Fig. 5C in that farms within the MiFAS boundary can also integrate their own farming enterprises in addition to cross-farm interconnectedness. This configuration more closely reflects the highly synergised landscapes in Martin et al. (2016).

Finally, Fig. 5E reflects a functionally diverse MiFAS boundary comprising of multiple interconnected farms that are also associated with commonly shared vertically integrated enterprises. These may include cooperatives for the acquisition of inputs (as much between farmers in the boundary and from the market) or the sales of outputs, bio-gas collection for fuel or heat, larger-scale processing, and so on. As such, this configuration provisions for internal supply chains within the boundary. This could also include the labelling, certification or standardisation of products originating from MiFAS. In essence, this configuration involves a high degree of vertical and horizontal integration and coordination, and numerous value chains though not without substantial management complexity.

4. Value chain implications & conclusions

With this review, we are the first to systematically synthesise the literature to identify the value chain implications of MiFAS. In this process we identified four major themes that each implicate specific aspects of MiFAS value chains. We have looked at how value is created within the farming system as a consequence of farming design and the arrangement of enterprises and actors within the MiFAS boundary. We have raised the impacts that the farming environment have on MiFAS and the triggers that support or disincentivise MiFAS by allowing them to create more or less value than competing farming systems. We have also identified how ecosystem services contribute to value creation in MIFAS and the extent to which they can be internalised by expanding value chain processes. Lastly, we have also seen how MiFAS – being diversified and lacking in economy of scale – struggle to integrate value-added processes in longer supply and value chains.

Broadly speaking, MiFAS present many opportunities for value creation. The interconnectedness resulting from synergising farming enterprises are undeniably vital for value creation, from the provision of ecosystem services reducing dependence on external inputs and promoting greater resilience to production and price risks, to improved land-use efficiency compared to specialised systems particularly in vulnerable and marginal agricultural lands. Collaboration and territorial interconnectedness also amplify these benefits. Additionally, ecosystem services created in MiFAS are the linchpin for socio-ecological services that more broadly contribute towards its sustainability, creating value for the environment and the rural economy.

However, we were alerted to the fact that MiFAS face significant challenges in their value chains. The need for additional capital and labour in a sector already facing tight margins may offset the beneficial cost-cutting from reduced input requirements. All the while higher productivity is not always assured due to resource competition. Furthermore, the opportunity cost of transitioning to MiFAS from specialised systems, ranging from lost efficiency to learning curves, and the need for long-term planning is costly in the short-term. Nor is it favourable when diversified holdings are more intensive to manage. The difficulties to internalise environmental and societal value-added when both policy and market-driven approaches to support them are scant negatively contribute to the economic sustainability of MiFAS. Ongoing rural challenges such as farm succession and abandonment, while similarly impacting all manners of farms, also harm MiFAS and potentially more so due to their greater need for labour.

We find very little concrete evidence that comprehensively evaluates the value chain implications of MiFAS. While this review has incorporated value chain evidence from research across the developed world (and much more can still be attested to beyond the scope of this review), the preeminent focus of research has been on the ecological and environmental effects of integrating farming enterprises, with some implications on farm productivity. Statements on comparative yields and outputs and some costs have contributed to our analysis of MiFAS, but concrete evidence of the implications of these are few. Although we did find research that produced bio-economic models for policy scenario-building (such as in Graves et al., 2007), the literature generally presented little cross-examination of the economic implications of MiFAS, and conclusions lacked external validation. Little economic modelling or empirical analyses were present that could necessarily explain how, how much, and what kind of value is being created by MiFAS and which compared and contrasted results between MiFAS or specialised farming systems. This is particularly egregious as we therefore found limited evidence to support any hypothesis that MiFAS are (or are not) economically sustainable at present.

We take full note that the value chain implications of MiFAS have hardly been studied. To this end, economic research – value chain and otherwise – must now be conducted to systematically and quantitatively assess MiFAS. By operationalising MiFAS to facilitate comparisons on the basis of system design, which, combined with the frameworks for assessing coordination and interconnectedness in MiFAS (Bell and Moore, 2012; Martin et al., 2016; Moraine et al., 2014), we have laid the groundwork for this assessment. Naturally, we expect that the value chains and resilience differences between MiFAS and in contrast to specialised systems should be assessed. For instance, research should focus on how to enhance MiFAS value chains and to more equitably internalise social and ecological services such that it can become more economically sustainable and compete in the wider food production environment. In turn, research should inform better farming strategies and designs, policy recommendations, and market-driven incentives for MiFAS adoption.

In conclusion, many of the issues raised in the introduction could be addressed by MiFAS. MiFAS have the potential to improve yields and land-use efficiency by improving the efficiency of resources such as nutrients and land, and particularly with greater inter-farmer partnerships and collaborations. At the same time, ecosystem services in MiFAS can create greater value for the agroecosystem. In spite of these, economic challenges at the farm-level, such as high opportunity costs to adopt MiFAS and high fixed and labour costs, can reduce the overall benefits of MiFAS. Yet, bridging the gap between fundamentally globalised challenges, pervasive in the agricultural systems of the developed world, and practically useful MiFAS requires more focus and a concerted effort to assess value chain implications. The reduction of mixed and agroforestry systems in the developed world (for instance in Europe, EUROSTAT, 2022) should serve as an indication that for such systems to prosper, the economic implications of agroecological systems must be considered with the same importance as ecosystem services. We fundamentally stress the importance of continuing to study MiFAS to quantitatively assess value chain implications and produce sound farmer, market, and policy recommendations.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my data/code at the Attach File step

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2023.103606.

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